

# General Market Impact Classifier

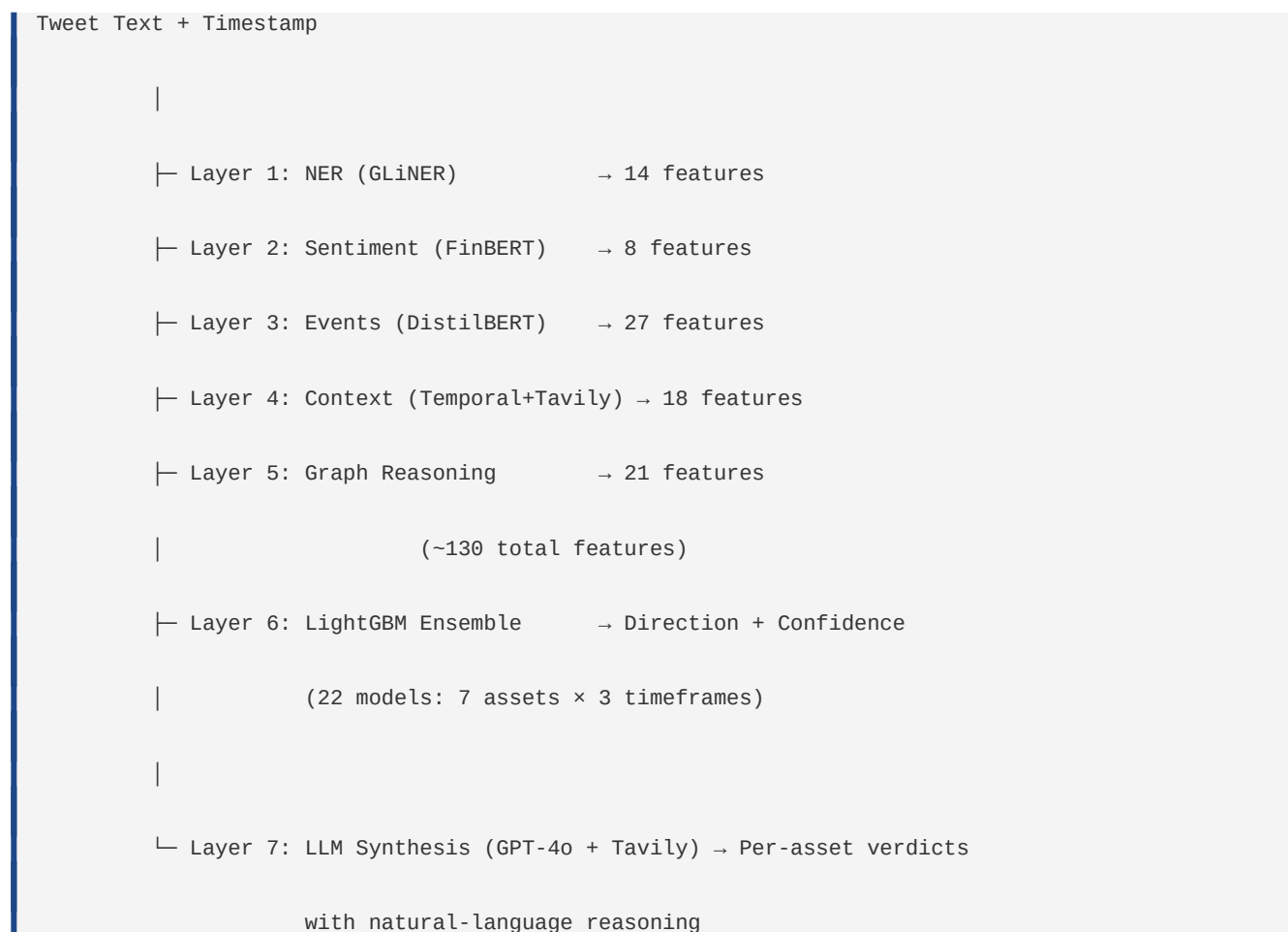
## Solution Overview

The **General Market Impact Classifier** is a 7-layer machine learning pipeline that predicts how political tweets impact seven global financial asset classes across three prediction horizons (1m, 5m, 10m).

**Architecture:** 5 NLP feature extraction layers + 1 LightGBM statistical ensemble + 1 LLM reasoning synthesis layer. Each layer is modular and independently debuggable.

**Output:** Per-asset market direction predictions (+1 bullish, 0 neutral, -1 bearish) with confidence scores and natural-language reasoning.

## System Architecture



## Layer Descriptions

**Layer 1 — NER (GLiNER):** Extracts persons, countries, commodities, organizations, and policy terms. Maps entities to target asset classes via curated lookup table. 14 features.

**Layer 2 — Sentiment (FinBERT):** Financial sentiment classifier (pre-trained on financial news).

Produces positive/negative/neutral probabilities and compound sentiment score. 8 features.

**Layer 3 — Event Detection (DistilBERT + Head):** Hybrid neural + rule-based classification across 13 event types (trade war, sanctions, conflict, monetary policy, election, regulation, energy crisis, etc.). 27 features.

**Layer 4 — Context Enrichment:** Temporal features (trading session, hour, day-of-week), stylistic signals (urgency, caps ratio, thread position), and **live market news from Tavily API** (required). 18 features.

**Layer 5 — Entity-Event-Asset Graph Reasoning:** Directed graph with 97 domain-knowledge transmission rules encoding how events propagate to asset classes. Signals modulated by sentiment. 21 features.

**Layer 6 — LightGBM Ensemble:** 1 binary relevance classifier + 21 direction classifiers (7 assets × 3 timeframes). 5-fold StratifiedKFold cross-validation, class-weight balanced.

**Layer 7 — LLM Reasoning (Azure OpenAI GPT-4o, Required):** Fetch real-time market news from Tavily → Assemble structured prompt with all layer outputs + LightGBM predictions + live news → Call GPT-4o → Per-asset direction + confidence + natural-language reasoning.

**Data & Training**

**Dataset:** 3,336 tweets with market direction labels for 7 assets at 3 timeframes. **Split:** Temporal (80% train = 2,669, 20% test = 667). **Training Time:** ~60 minutes on CPU (15 min with GPU). **Inference Latency:** ~2,000–5,000ms per tweet.

**Results**

Relevance Detection (Layer 6, LightGBM, 667 test samples)

| Metric        | Value   |
|---------------|---------|
| Accuracy      | 85.31 % |
| ROC-AUC       | 89.72 % |
| Recall        | 75.35 % |
| F1 (Weighted) | 85.14 % |

CV Accuracy     95.47%

The relevance gate is the highest-confidence component, correctly identifying market-moving tweets with strong precision and recall.

Direction Prediction (Layer 6, LightGBM, 5m horizon)

Mean Across All Assets: Accuracy: 44.14% | Balanced Accuracy: 38.08% | F1 (Macro): 37.50%

| Asset       | Accuracy | Balanced Acc | F1 (Macro) | MCC     |
|-------------|----------|--------------|------------|---------|
| wheat       | 53.97%   | 50.30%       | 49.19%     | 0.2925  |
| btc         | 46.93%   | 34.36%       | 33.29%     | 0.0178  |
| equities    | 42.58%   | 34.27%       | 33.99%     | 0.0057  |
| gold        | 43.63%   | 33.29%       | 32.88%     | -0.0154 |
| eurodollar  | 43.93%   | 41.74%       | 40.78%     | 0.0977  |
| cl          | 40.03%   | 37.06%       | 36.57%     | 0.0303  |
| treasury_2y | 35.23%   | 34.58%       | 34.55%     | 0.0194  |

**Insight:** Wheat shows the strongest signal (~54% accuracy) due to direct geopolitical/agricultural sensitivity. Directional prediction remains challenging due to market microstructure noise and complex macro drivers.

Layer 7 LLM Reasoning (GPT-4o, 5m horizon)

| Component               | GPT-4o | LightGBM |
|-------------------------|--------|----------|
| Mean Direction Accuracy | 22.04% | 44.61%   |
| Mean F1 (Macro)         | 14.10% | 34.92%   |
| Relevance Accuracy      | 85.61% | 86.96%   |

Per-Asset Agreement (LLM vs LightGBM): wheat 57.7% | treasury\_2y 41.4% | eurodollar 15.9% | cl

12.1% | equities 10.2% | gold 7.1% | btc 6.3%

**Insight:** GPT-4o serves as a **reasoning synthesis layer** that produces interpretable explanations, grounds predictions in live market context (Tavily), and achieves parity on relevance classification (85.6%). The LLM layer's value is complementary interpretability and context integration.